

Conformal Transformation Laws

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Introduction

We study the transformation laws for geometric quantities under a conformal change of metric. Let (M^n, g) be an n -dimensional Riemannian manifold, and consider the conformally related metric

$$\tilde{g} = e^{2f}g,$$

where $f \in C^\infty(M)$.

The Riemann curvature tensor, Ricci curvature tensor, and scalar curvature associated with g will be denoted by Rm , Ric , and Sc , respectively. Their counterparts with respect to $\tilde{g}$ will be denoted by $\widetilde{\text{Rm}}$, $\widetilde{\text{Ric}}$, and $\widetilde{\text{Sc}}$.

Now let $\Sigma^{n-1} \subset M^n$ be an immersed hypersurface. With respect to g , denote its unit normal, second fundamental form, and mean curvature by ν , A , and H , respectively. The corresponding quantities computed with respect to $\tilde{g}$ will be denoted by $\tilde{\nu}$, $\tilde{A}$, and $\tilde{H}$.

Our goal is to derive the transformation laws for these intrinsic and extrinsic geometric quantities under the conformal change

$$g \mapsto \tilde{g} = e^{2f}g.$$

Geometric Quantities and Conventions

In this section, we fix the notation and sign conventions used throughout the note.

Levi-Civita Connection

Let ∇ and $\tilde{\nabla}$ denote the Levi-Civita connections of g and $\tilde{g}$, respectively. The gradient, Hessian, and Laplace–Beltrami operator associated with g are defined by

$$g(\nabla f, X) = df(X),$$

$$\nabla^2 f(X, Y) = X(Yf) - (\nabla_X Y)f,$$

and

$$\Delta f = \text{tr}_g \nabla^2 f.$$

Curvature Tensors

We define the Riemann curvature operator by

$$\text{Rm}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z.$$

Its $(0, 4)$ -tensor version is

$$\text{Rm}(X, Y, Z, W) = g(\text{Rm}(X, Y)Z, W).$$

The Ricci and scalar curvatures are obtained by taking traces:

$$\text{Ric}_{jl} = g^{ik} \text{Rm}_{ijkl}, \quad \text{Sc} = g^{jl} \text{Ric}_{jl}.$$

For symmetric $(0, 2)$ -tensors P and Q , we use the Kulkarni–Nomizu product

$$\begin{aligned} (P \oslash Q)(X, Y, Z, W) &= P(X, Z)Q(Y, W) + P(Y, W)Q(X, Z) \\ &\quad - P(X, W)Q(Y, Z) - P(Y, Z)Q(X, W). \end{aligned}$$

Hypersurface Geometry

Let ν be a choice of unit normal along Σ . For tangent vector fields $X, Y \in T\Sigma$, we define the second fundamental form by

$$A(X, Y) = g(\nabla_X \nu, Y).$$

The mean curvature is the trace of A with respect to the induced metric:

$$H = \text{tr}_{g|_{T\Sigma}} A.$$

Sign conventions

The signs in the formulas depend on the conventions for Rm , A , and Δ . All formulas below use the conventions fixed above. In particular, H is the trace of A , rather than its average.

Conformal Transformation Formulas

Define the symmetric $(0, 2)$ -tensor

$$B = \nabla^2 f - df \otimes df + \frac{1}{2} |\nabla f|^2 g.$$

With the conventions above, the conformal transformation laws are as follows.

Levi-Civita Connection

For vector fields X, Y on M ,

$$\widetilde{\nabla}_X Y = \nabla_X Y + X(f)Y + Y(f)X - g(X, Y)\nabla f.$$

Riemann Curvature Tensor

As a $(0, 4)$ -tensor,

$$\widetilde{\text{Rm}} = e^{2f} (\text{Rm} - g \oslash B).$$

Ricci Curvature

The Ricci curvature transforms according to

$$\widetilde{\text{Ric}} = \text{Ric} - (n-2)(\nabla^2 f - df \otimes df) - (\Delta f + (n-2)|\nabla f|^2)g.$$

Scalar Curvature

The scalar curvature satisfies

$$\widetilde{\text{Sc}} = e^{-2f} [\text{Sc} - 2(n-1)\Delta f - (n-1)(n-2)|\nabla f|^2].$$

Hypersurface Quantities

The unit normal, second fundamental form, and mean curvature transform as

$$\tilde{\nu} = e^{-f}\nu,$$

$$\widetilde{A}(X, Y) = e^f (A(X, Y) + \nu(f)g(X, Y)),$$

and

$$\widetilde{H} = e^{-f}(H + (n-1)\nu(f)).$$

Derivations

Difference of the Levi-Civita Connections

Set

$$D(X, Y) = \widetilde{\nabla}_X Y - \nabla_X Y.$$

Since both connections are torsion-free, D is symmetric in X and Y . Applying the Koszul formula to $\tilde{g} = e^{2f}g$ gives

$$\begin{aligned} 2\tilde{g}(\widetilde{\nabla}_X Y, Z) &= X\tilde{g}(Y, Z) + Y\tilde{g}(X, Z) - Z\tilde{g}(X, Y) \\ &\quad + \tilde{g}([X, Y], Z) - \tilde{g}([X, Z], Y) - \tilde{g}([Y, Z], X). \end{aligned}$$

Using

$$X(\tilde{g}(Y, Z)) = e^{2f}(2X(f)g(Y, Z) + X(g(Y, Z)))$$

and comparing with the Koszul formula for g , we obtain

$$g(D(X, Y), Z) = X(f)g(Y, Z) + Y(f)g(X, Z) - Z(f)g(X, Y).$$

Because $Z(f) = g(\nabla f, Z)$, this yields

$$D(X, Y) = X(f)Y + Y(f)X - g(X, Y)\nabla f.$$

Therefore,

$$\widetilde{\nabla}_X Y = \nabla_X Y + X(f)Y + Y(f)X - g(X, Y)\nabla f.$$

Transformation of the Riemann Curvature Tensor

For two torsion-free connections related by $\widetilde{\nabla} = \nabla + D$, their curvature operators satisfy

$$\begin{aligned} \widetilde{\text{Rm}}(X, Y)Z &= \text{Rm}(X, Y)Z + (\nabla_X D)(Y, Z) - (\nabla_Y D)(X, Z) \\ &\quad + D(X, D(Y, Z)) - D(Y, D(X, Z)). \end{aligned}$$

Substituting

$$D(X, Y) = X(f)Y + Y(f)X - g(X, Y)\nabla f$$

and collecting the Hessian and gradient terms gives

$$e^{-2f}\widetilde{\text{Rm}} = \text{Rm} - g \oslash \left(\nabla^2 f - df \otimes df + \frac{1}{2}|\nabla f|^2 g \right).$$

Equivalently,

$$\widetilde{\text{Rm}} = e^{2f}(\text{Rm} - g \oslash B).$$

Transformation of the Ricci Curvature

Contracting the first and third indices of $g \oslash B$ gives

$$g^{ik}(g \oslash B)_{ijkl} = (n-2)B_{jl} + (\text{tr}_g B)g_{jl}.$$

Moreover,

$$\text{tr}_g B = \Delta f + \frac{n-2}{2}|\nabla f|^2.$$

Since $\tilde{g}^{ik} = e^{-2f}g^{ik}$, tracing the transformed Riemann tensor yields

$$\begin{aligned}\widetilde{\text{Ric}} &= \text{Ric} - (n-2)B - (\text{tr}_g B)g \\ &= \text{Ric} - (n-2)(\nabla^2 f - df \otimes df) - (\Delta f + (n-2)|\nabla f|^2)g.\end{aligned}$$

Transformation of the Scalar Curvature

Taking the trace of $\widetilde{\text{Ric}}$ with respect to $\tilde{g}$, we find

$$\widetilde{\text{Sc}} = \tilde{g}^{ij}\widetilde{\text{Ric}}_{ij} = e^{-2f}g^{ij}\widetilde{\text{Ric}}_{ij}.$$

The g -trace of the Ricci transformation formula is

$$g^{ij}\widetilde{\text{Ric}}_{ij} = \text{Sc} - 2(n-1)\Delta f - (n-1)(n-2)|\nabla f|^2.$$

Consequently,

$$\widetilde{\text{Sc}} = e^{-2f} [\text{Sc} - 2(n-1)\Delta f - (n-1)(n-2)|\nabla f|^2].$$

Transformation of the Unit Normal

Because ν is a g -unit normal,

$$g(\nu, \nu) = 1.$$

The vector field $e^{-f}\nu$ remains normal to Σ , and

$$\tilde{g}(e^{-f}\nu, e^{-f}\nu) = e^{2f}g(e^{-f}\nu, e^{-f}\nu) = 1.$$

Hence

$$\tilde{\nu} = e^{-f}\nu.$$

Transformation of the Second Fundamental Form

For $X, Y \in T\Sigma$,

$$\tilde{A}(X, Y) = \tilde{g}(\widetilde{\nabla}_X \tilde{\nu}, Y).$$

Using $\tilde{\nu} = e^{-f}\nu$, we compute

$$\widetilde{\nabla}_X \tilde{\nu} = -e^{-f}X(f)\nu + e^{-f}\widetilde{\nabla}_X \nu.$$

The connection transformation formula gives

$$\tilde{\nabla}_X \nu = \nabla_X \nu + X(f)\nu + \nu(f)X,$$

because $g(X, \nu) = 0$. Therefore,

$$\tilde{\nabla}_X \tilde{\nu} = e^{-f}(\nabla_X \nu + \nu(f)X).$$

It follows that

$$\begin{aligned} \tilde{A}(X, Y) &= e^{2f}g(e^{-f}(\nabla_X \nu + \nu(f)X), Y) \\ &= e^f(A(X, Y) + \nu(f)g(X, Y)). \end{aligned}$$

Transformation of the Mean Curvature

Let $\{e_1, \dots, e_{n-1}\}$ be a local g -orthonormal frame tangent to Σ . Then $\{e^{-f}e_1, \dots, e^{-f}e_{n-1}\}$ is $\tilde{g}$ -orthonormal. Hence

$$\begin{aligned} \tilde{H} &= \sum_{a=1}^{n-1} \tilde{A}(e^{-f}e_a, e^{-f}e_a) \\ &= e^{-f} \sum_{a=1}^{n-1} (A(e_a, e_a) + \nu(f)) \\ &= e^{-f}(H + (n-1)\nu(f)). \end{aligned}$$

Thus,

$$\tilde{H} = e^{-f}(H + (n-1)\nu(f)).$$

← Back to all notes